MEDICATION SAFETY SIGNAL MONITOR · HAND-IN R04

The application

Version 1.2, 6 November 2026: the safety officer's morning in Workshop, click by click, and how it was built

The question this report answers

How does a hospital pharmacy safety officer use the app each morning, what does each number mean, and how was the app built (and how is it changed) in Palantir Workshop?

One-line answer

One Workshop module with 4 pages behind header tabs: a Morning brief with three numbers and the as-of date of each source, then one page per question. On the first question she can open the exact label versions that were searched, and record a triage decision that is written back to the Ontology and survives every rebuild.

What this is	What this is not
• A click-by-click guide to the officer's morning, page 0 to page 3, with screenshots of the live app • A plain explanation of every number on the screen, and what it does not mean • A guide to the widgets on the screen and how an editor changes them (Edit mode, variables, widget config) • An honest list of the gaps and the next UI improvements	• A user study: the persona is fictional. Load and page-switch times were measured by a script; a human two-minute walk-through has not been run • The data model behind the screen (see R03, the Ontology) • Proof of drug harm: a signal is a reason to look, not a finding • Reviewed by a pharmacist

How to read this. Chapter 1 is the one-page summary, with the words you need and what the app means for the officer. Chapter 2 introduces her and her three questions. Chapter 3 is the user guide: the morning routine, click by click, with screenshots. Chapter 4 explains every number. Chapter 5 is for editors: the widgets, the variables, and how to change them. Chapter 6 shows what we changed for usability and why. Chapter 7 lists the gaps and the next UI improvements. Each chapter ends with a gate box: what we required, what we found, and what to do if it fails.

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01

Summary in plain language

Who and why. A hospital pharmacy safety officer has about fifteen minutes each morning for safety review (design-thinking canvas; an assumption, not a measurement). She asks three questions: (1) which drugs have side effects being reported that the label does not mention, (2) which drugs have an active recall, (3) what has changed since yesterday. She trusts a flag only if she can see its source.

Which drugs. Her formulary in this pilot is 13 drugs: 8 opioids (oxycodone, hydrocodone, morphine, fentanyl, hydromorphone, methadone, buprenorphine, tramadol) and 5 GLP-1 medicines (semaglutide, liraglutide, dulaglutide, tirzepatide, exenatide).

Words used in this report

Word	Meaning
Foundry	Palantir's data platform. Our data, pipelines, data model and app all live in one Foundry project.
Ontology, object, link	Foundry's business layer on top of the data tables. An *object* is one thing (a drug, a recall, a signal); a *link* connects two objects (a signal to its drug).
Action	A form that writes to the Ontology. Ours creates a *Triage Decision* object.
Workshop	Foundry's app builder. An app is a *module*; it shows objects in widgets.
FAERS	The FDA Adverse Event Reporting System: public reports of suspected side effects, released quarterly.
PRR	Proportional reporting ratio: how much more often a reaction is reported with this drug than with all other drugs.
Reference label	The official US label text, from DailyMed (the National Library of Medicine's label library). One reference label version per product.
Signal	A drug and reaction pair that is reported unusually often (the Evans rule: 3+ reports, PRR 2+, chi-squared 4+; D006) and is not found in the drug's reference labels, or only possibly in label (108: found only through a medium-band synonym).
Ranked signal	A signal in the review queue: not in label, a clinical reaction (not, for example, an efficacy complaint), and not mostly from mass reports.

What she gets

A Workshop module called **MSSM Safety Officer**, published as v0.10.0. It has 4 pages, switched by tabs in the header:

Page	Answers	What she sees on 6 November 2026
0 · Morning brief	All three, at a glance	1,073 ranked signals · 126 active recalls · 26 changes in the last 30 days, plus an as-of date for each source
1 · Not in label	Question 1	Ranked signals per drug; the evidence record; the label versions that were searched, with DailyMed links; what the AI read in those labels, quoted word for word; the button *Triage this signal*
2 · Recalls	Question 2	900 recalls for our drugs, filterable to active; detail of the selected recall
3 · What changed	Question 3	Dated changes to labels and recalls (and signals, after a data refresh); a date range means 'since my last review'

The write-back. On page 1 a user selects a signal and clicks *Triage this signal*. They pick a decision (reviewing, dismissed, escalated, monitoring) and write a reason. Foundry stores it as a new *Triage Decision* object, with the user, the time, and the exact computation that was reviewed. It is still there after a full rebuild of the data TESTED.

Read the first number this way

The 1,073 ranked signals are a review queue, not 1,073 findings. A signal means a reaction is reported unusually often with this drug and was not found in its label. It does not prove the drug causes it.

What this means for Dana (the safety officer)

Fast to open. Measured by a script in View mode: the Morning brief is fully loaded in 11.5 s (including Foundry start-up), and each page switch takes 0.8 to 2.4 s (timed on v0.9.0; v0.10.0 only adds one table below the fold on page 1, not re-timed). Her reading time was not measured.

Card 3 is not 'since yesterday'. It counts changes in the last 30 days. For 'since my last review' she sets the start date on page 3.

Signals move once a quarter. New signals appear only when a new FAERS quarter arrives, and FAERS is about 130 days behind. Day to day, page 3 shows new label versions and recalls.

She can check 'not in label' herself. For NAION she sees the 3 label versions that were searched and can open each in DailyMed. Below them she sees what the AI read in those labels, quoted word for word: diabetic retinopathy is there, no optic-nerve term is. So exact text, synonyms and the AI agree NAION is 'not in label'. NAION is a *known positive*, not a discovery: EMA PRAC (6 Jun 2025) and a WHO alert (27 Jun 2025) named it, and our 3 US labels do not mention it (DECISIONS learning 2026-11-05).

She cannot triage yet. Only the project owner can submit a Triage Decision today. Giving her role that permission is a production step.

Label	Meaning
OBSERVED	Read directly from a public source (FAERS, DailyMed, openFDA, RxNorm), with the query that produced it.
DERIVED	Computed by our pipeline from observed data, for example a PRR or a count per drug and reaction.
ASSUMED	Not verified: taken as true for now because the data or the brief is silent. It could be wrong.
DESIGNED	A design choice we made and wrote down (the decision log says why). It may be built; the status band says how far.
TESTED	Built, run, and checked against evidence.
PLANNED	On the plan; not built yet.
PROPOSED	Suggested; waiting for a decision.

Gate: Summary TESTED

Required evidence: All three questions answered from one module; one decision written back and kept.

Result: Met on 5 November 2026 (baseline gate) and again on 6 November (v0.10.0). Machine timing done on v0.9.0 (brief 11.5 s, switches 0.8 to 2.4 s). A human two-minute walk-through has not been run.

If it fails: If a question needs another tool, the app has not done its job: add it to a page.

02

The officer and her three questions

We designed for one person. Every page had to pass her test: does it answer one of her questions, and can she see where the answer came from?

We used *design thinking* (Empathize, Define, Ideate, Prototype, Test). With no real officer to interview. Problem: she needs a fast, trustworthy morning view of drug risks, because today she learns of them late and cannot defend an escalation without evidence. The persona is **fictional**, built from the brief ASSUMED.

Persona	Dana, medication safety officer, PharmD (Doctor of Pharmacy), in a large health system (fictional)
Says	"I found out about that recall from a nurse." · "I don't have time to read FAERS."
Thinks	"Am I missing something right now?" · "If I escalate a false alarm, I lose credibility."
Does	Scans FDA emails, MedWatch alerts (the FDA's safety-alert service) and newsletters; brings issues to the safety committee.
Feels / trust bar	Anxious about being late; sceptical of black boxes. "Show me where it says that."

From 'how might we' to a page

Status here means built and checked by us in View mode (the user's view). No real officer has tested it.

How might we ...	What we built	Status
show only what is new or worse?	Page 3 (What changed) with a date range; the Morning brief counts the last 30 days	TESTED
make 'not on the label' provable in one click?	Page 1: label status, check method, and the panel *Reference labels searched* (version, effective date, sections, DailyMed link)	TESTED
show how confident we are without a statistics lecture?	Plain labels: 'PRR (x more often than other drugs)', CI low, caution flags	TESTED
let her record a decision?	Triage button, decision list under the signal (owner-only today)	TESTED
show her what the AI read in the label?	Page 1 table *What the AI read in these labels*: reaction, section, word-for-word quote, label (v0.10.0)	TESTED
warn her when the AI might be wrong?	Not on screen yet. Measured: 745 of 745 AI quotes found in the label text; monitoring checks it each build (D046)	PLANNED

Her test, and where we stand

Test (from our design canvas)	Pass if	Result
Morning in two minutes	She answers all three questions in two minutes or less	Machine side (timed on v0.9.0): brief loads in 11.5 s, page switches 0.8 to 2.4 s OBSERVED. Human walk-through not run PLANNED
Trust check	Every flag reaches its source in two clicks or less	Met on page 1: click a signal for the evidence and the labels searched; click again to open the label in DailyMed TESTED. The FAERS case reports behind a signal are reachable by query, not yet from her screen

Gate: User fit TESTED

Required evidence: Each page maps to one of the three questions; the trust rule is visible on screen.

Result: Met: four pages, one per question plus the brief; the labels-searched panel puts the trust rule on page 1. Not validated with a real officer.

If it fails: A page that answers no question should be removed, not polished.

03

The morning routine, click by click

This chapter is the user guide. Four stops, left to right across the header tabs. The 20-minute demo follows the same path.

Step 0 · Open the app

1. In Foundry, open the project folder `Workshop/` and click the module *MSSM Safety Officer* (its header reads 'MSSM').
2. It opens on the tab *Morning brief* in **View mode**, the mode every user sees. (Editors switch to Edit mode; see chapter 5.)

Step 1 · Page 0, Morning brief: three numbers and 'data as of'

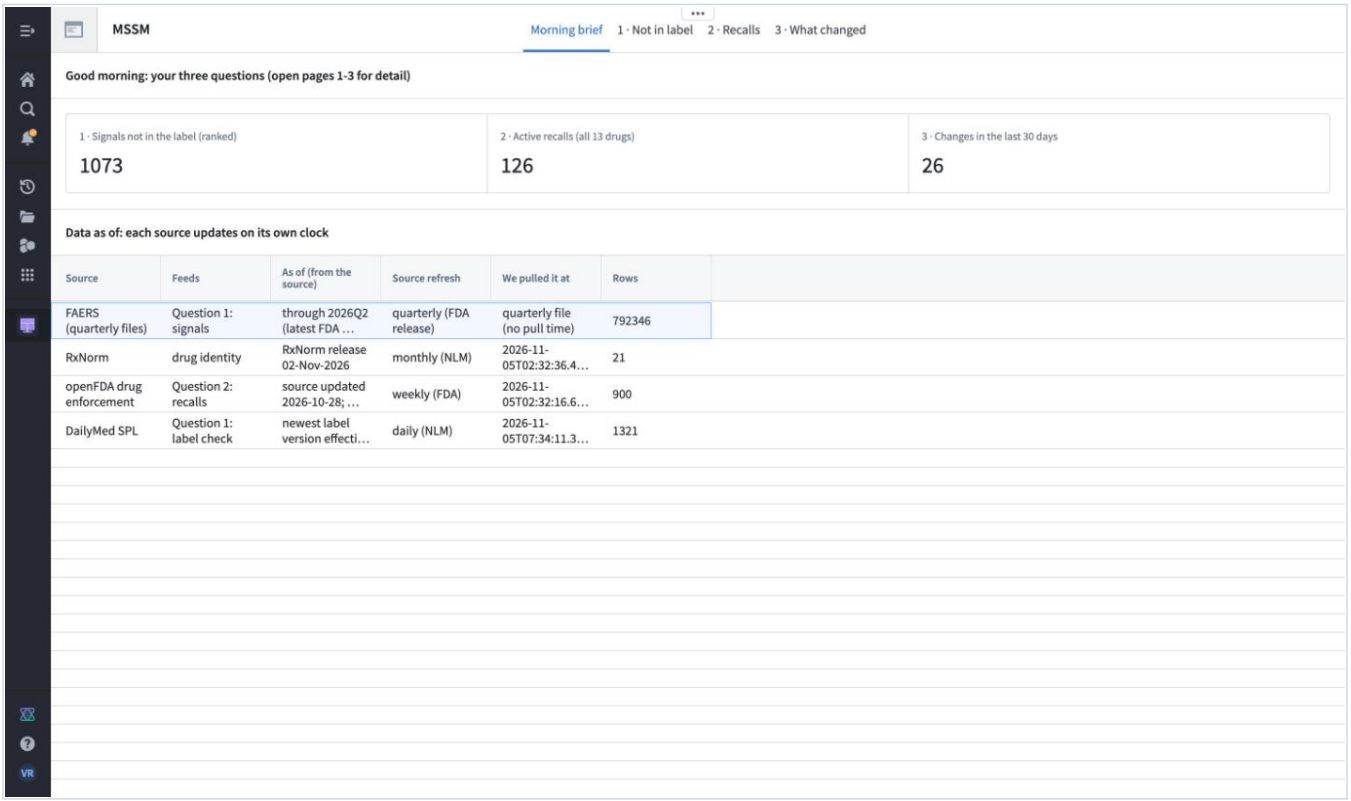


Figure 1. Page 0, Morning brief. Three metric cards answer the three questions; the table below says how fresh each source is. FAERS shows 'quarterly file (no pull time)' on purpose.
Source: docs/11-handin/screens/app-0-morning-brief.jpg (Workshop, 2026-11-05)

1. Read card 1: 1,073 ranked signals not in the label. This is the review queue for question 1.
2. Read card 2: 126 active recalls across all 13 drugs. Active means the openFDA status is *Ongoing*.
3. Read card 3: 26 changes in the last 30 days. It is a 30-day window, not 'since yesterday'; page 3 narrows it to her last review.
4. Read the table *Data as of*. Each source updates on its own clock: FAERS quarterly (here through 2026Q2), openFDA recalls weekly, DailyMed labels daily, RxNorm monthly. *We pulled it at* shows when our pipeline fetched RxNorm, openFDA and DailyMed. For FAERS it says 'quarterly file (no pull time)' on purpose: we load the FDA's quarterly files, so the useful date is the quarter, not a clock time (`ontology_changes.py` l.152). These are the sources' own clocks; our pipeline refreshes only when a build runs, and there is no schedule yet (another month). A 'nothing changed' is only as fresh as the oldest source.
5. Click a tab in the header to go deeper.

Step 2 · Page 1, Not in label: pick a drug and read the evidence

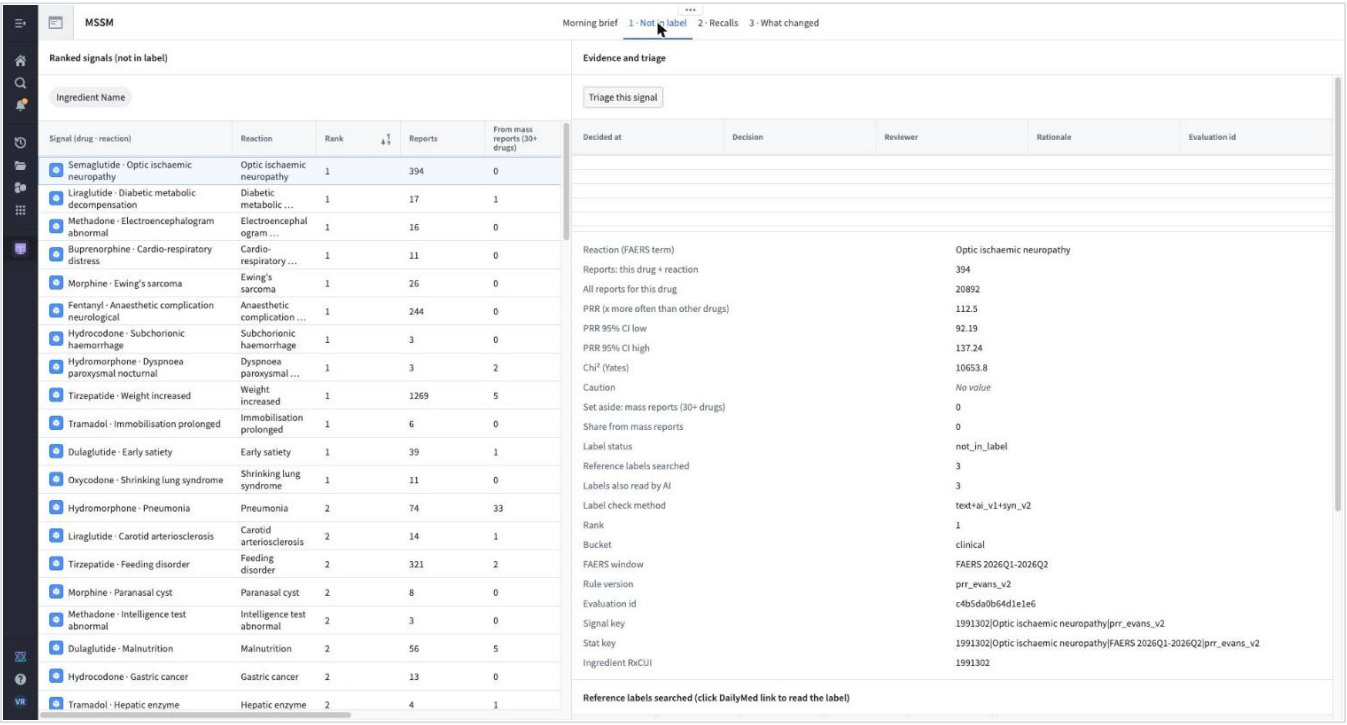


Figure 2. Page 1 with semaglutide's top signal selected: NAION (FAERS term 'Optic ischaemic neuropathy'). Left: the drug filter and the ranked signals. Right: the triage button, past decisions and the signal record, with *Labels also read by AI* = 3 and evaluation id c4b5da0b64d1e1e6.

Source: docs/11-handin/screens/app-1d-naion-ai-read-3.jpg (Workshop v0.10.0, 2026-11-06)

1. Click the tab *1 · Not in label*.
2. Click the pill *Ingredient Name* (the drug filter). It lists the drugs with the number of ranked signals each has. Tick semaglutide. The table now shows only that drug.
3. The table is sorted by rank within each drug. Rank 1 for semaglutide is *Optic ischaemic neuropathy* (NAION, a sudden loss of vision in one eye).
4. Click the row. The right side, *Evidence and triage*, shows the signal record: 394 reports, PRR 112.5 (reported about that many times more often than with other drugs), lower 95% bound 92.19, label status *not_in_label*, *Reference labels searched* 3, *Labels also read by AI* 3, and the label check method.

Step 3 · Page 1: open the labels that were searched

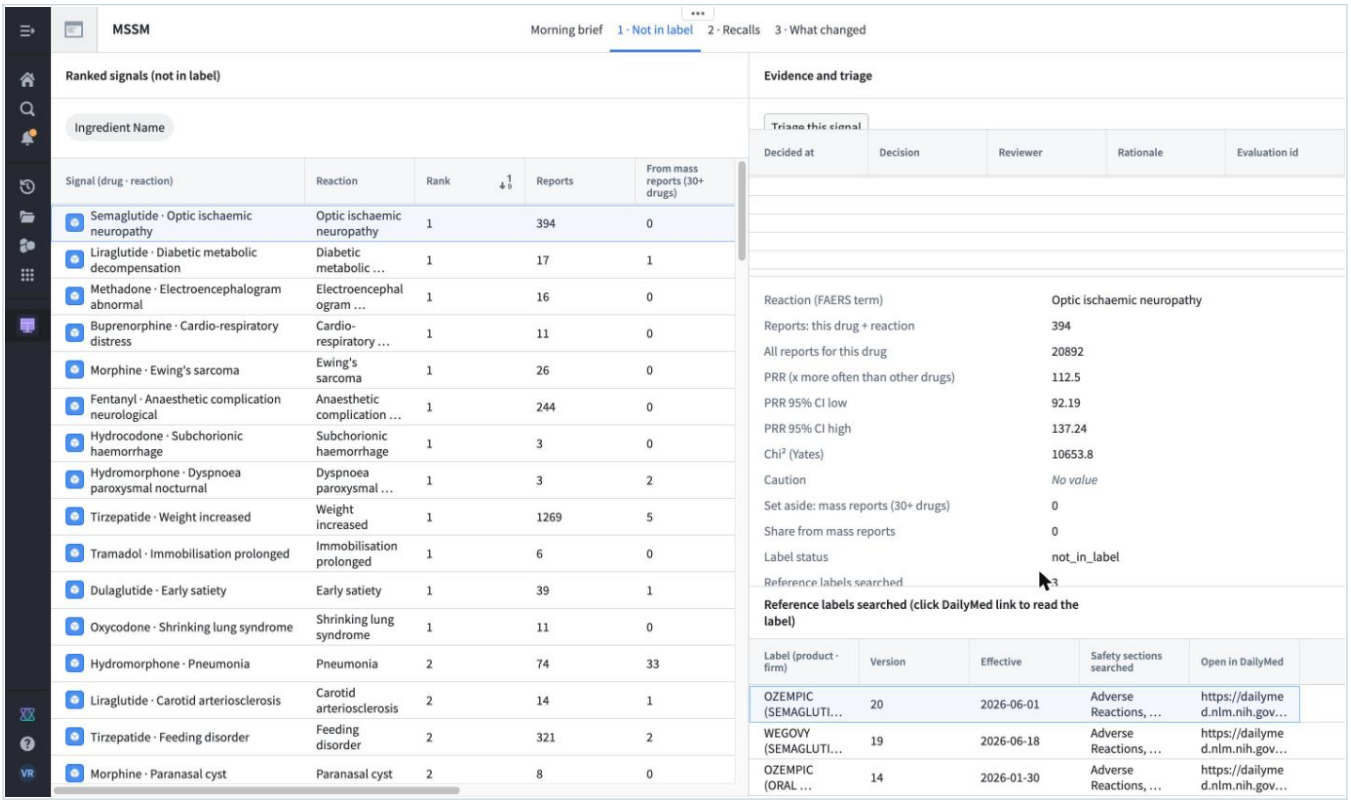


Figure 3. Page 1, lower right: the panel 'Reference labels searched' for NAION. Three semaglutide label versions, each with its effective date, the safety sections searched and a DailyMed link.
Source: docs/11-handin/screens/app-1c-labels-searched-naion.jpg (Workshop, 2026-11-05)

1. Scroll the right side down to the panel *Reference labels searched*. For NAION it lists 3 label versions: Ozempic v20 (effective 2026-06-01), Wegovy v19 (2026-06-18) and oral Ozempic v14 (2026-01-30) (A016; screenshot above).
2. Read the columns: label (product and firm), version, effective date, safety sections searched, and *Open in DailyMed*.
3. Click a DailyMed link. The exact label version opens on the DailyMed site. This is the trust rule on screen: the flag says 'not found in these three labels', and she can check it herself.

1. Back in the app, keep scrolling the right side to the table *What the AI read in these labels (every line is quoted word for word from the label)*. Columns: reaction, section, quote, label key (297 rows for semaglutide).
2. Look for the eyes: the AI found diabetic-retinopathy terms, and no optic-nerve or NAION term. That agrees with the flag: the label talks about the retina, not the optic nerve.

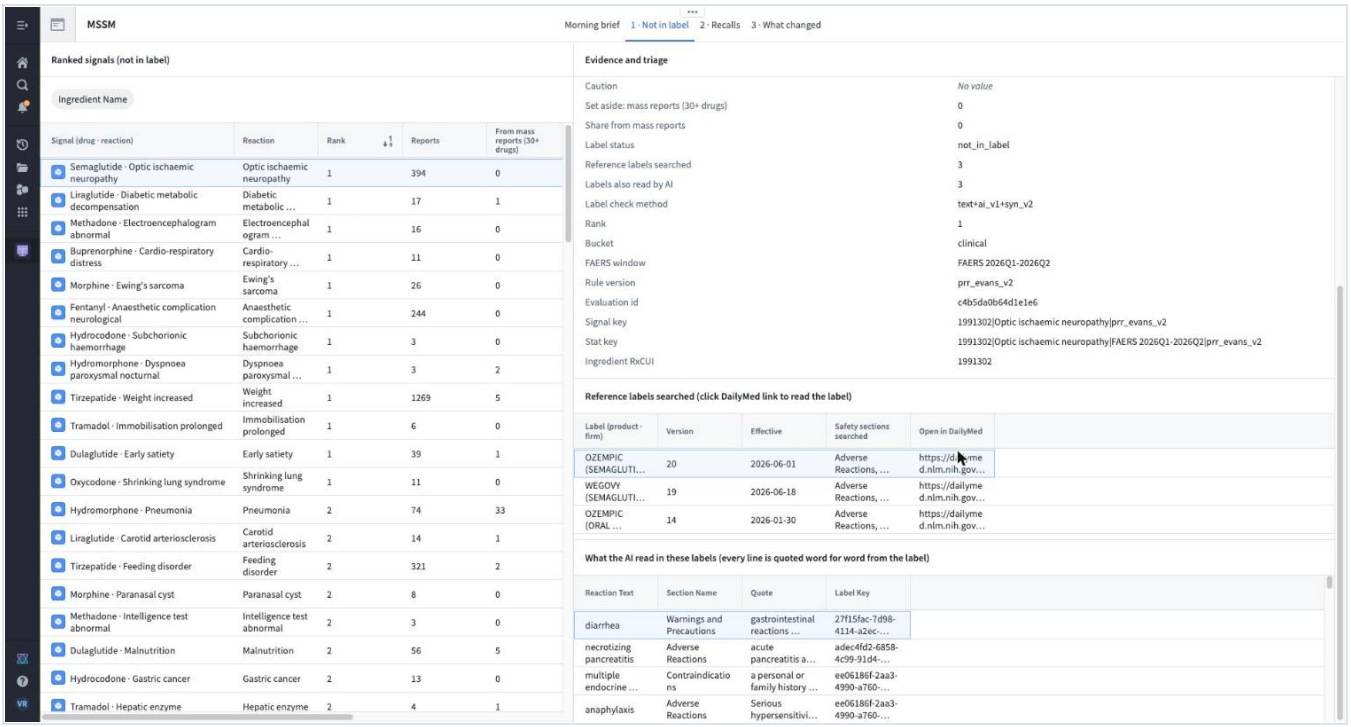


Figure 4. Page 1, right side scrolled down: *Reference labels searched* (3 labels with DailyMed links) and, below it, *What the AI read in these labels*: reaction · section · word-for-word quote · label key.
Source: docs/11-handin/screens/app-1e-ai-quotes-panel-naion.jpg (Workshop v0.10.0, 2026-11-06)

Behind the panels

How the panels find the labels: they follow existing Ontology links from the selected signal, Signal → Drug → Products → Reference Label, and then Label → Labeled Reactions for the AI table (A016, A019). There is no dedicated Signal → Label link type yet. *Labels also read by AI* is 3 here because the AI reads the 3 semaglutide labels since D041 v2 (6 Nov). The AI table shows every reaction the AI found in those labels, not only the ones that match a signal.

Step 4 · Page 1: record a triage decision

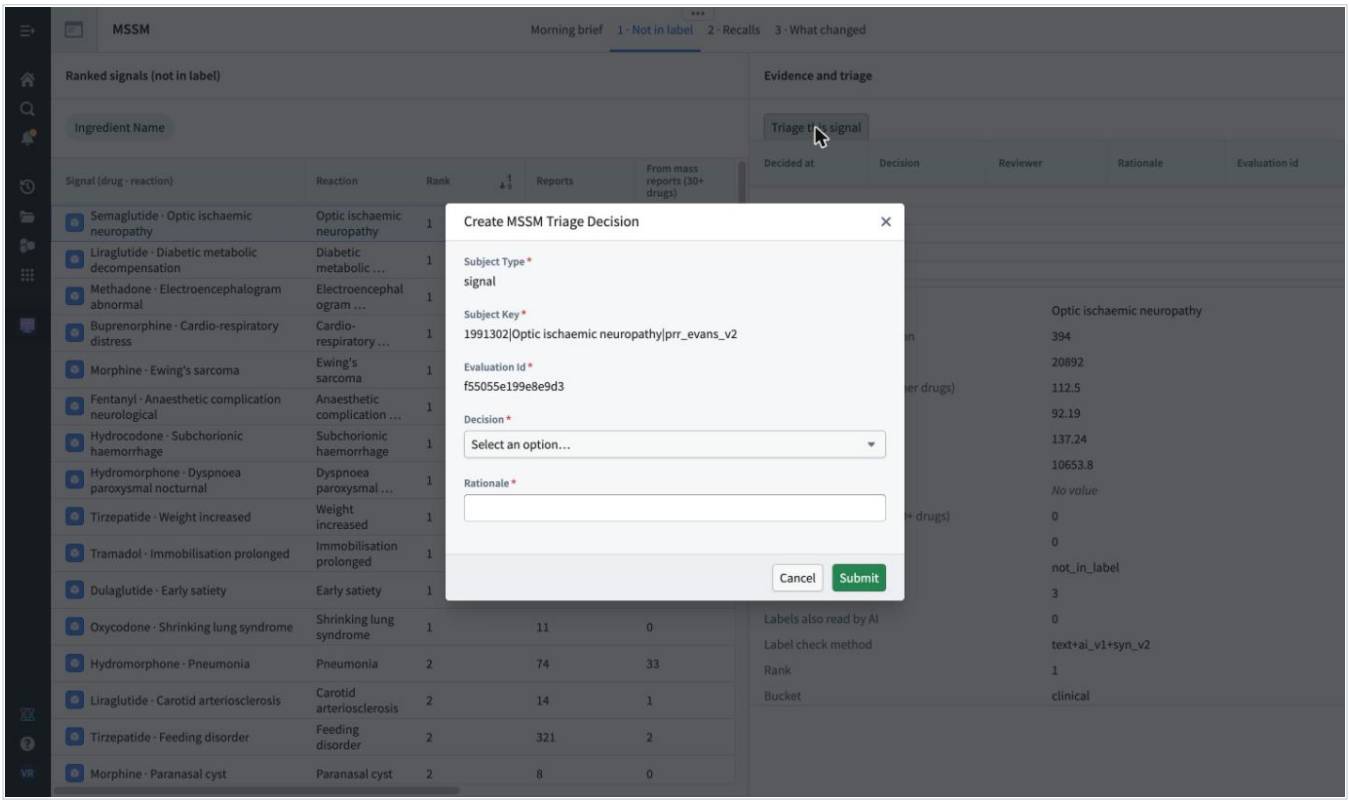


Figure 5. The form 'Create MSSM Triage Decision'. Subject Type, Subject Key and Evaluation Id are filled from the selected signal and locked. (This capture was cancelled, so nothing was saved.)
Source: docs/11-handin/screens/app-1b-triage-action-form.jpg (Workshop, 2026-11-05)

1. With the signal still selected, click *Triage this signal* (top of the right side).
2. The form opens. Subject Type ('signal'), Subject Key (drug, reaction and rule version) and Evaluation Id (a fingerprint of the numbers she is looking at) are filled in and cannot be changed.
3. Pick a *Decision*: reviewing, dismissed, escalated or monitoring. Write a *Rationale*. Click *Submit*. (*Cancel* saves nothing.)
4. The decision appears in the table above the signal record: decided at, decision, reviewer, rationale, evaluation id. The system sets the reviewer and the time.

> **Watch out**
> Today only the project owner can submit this Action (`foundry/ontology/README.md`, Action types). The officer's role needs that permission before a pilot with real users.

Step 5 · Page 2, Recalls

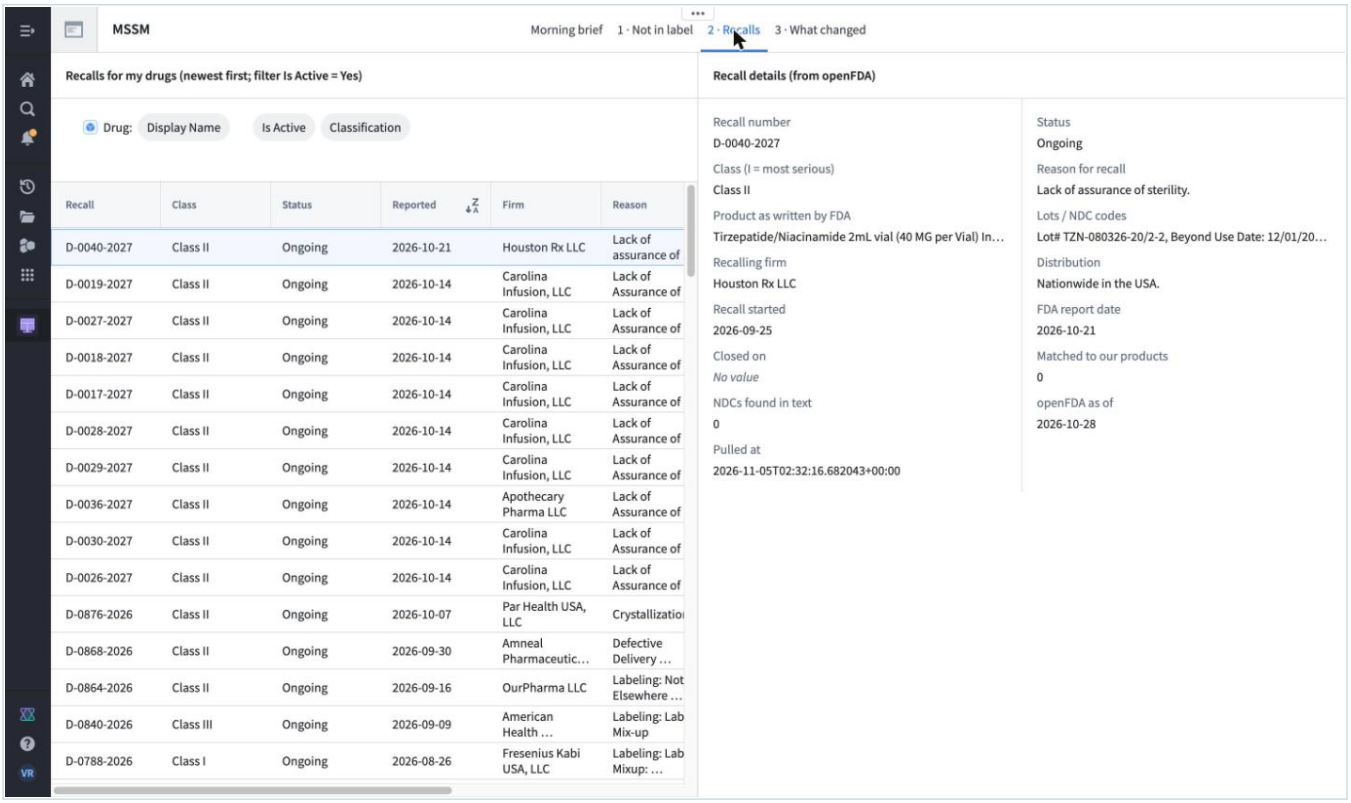


Figure 6. Page 2, Recalls, newest first. The selected recall, D-0040-2027, is the newest one at the baseline gate.
Source: docs/11-handin/screens/app-2-recalls.jpg (Workshop, 2026-11-05)

1. Click the tab *2 · Recalls*.
2. Use the three filter pills: *Drug: Display Name* (works through the Recall-to-Drug link, so a recall that names two drugs shows under both), *Is Active* (Yes = status Ongoing), *Classification* (Class I is the most serious, then II, then III).
3. The table is sorted by FDA report date, newest first. Click a recall to see 15 fields in reading order on the right (`foundry/ontology/README.md`): product as written by the FDA, firm, reason, lots, dates, and the openFDA as-of date.
4. Read *Matched to our products* with care. The newest recall, D-0040-2027 (reported 2026-10-21), is a compounded tirzepatide product. It names our ingredient but matches none of our products, so it shows 0. That is correct, not a bug, but the screen does not yet say so.

Step 6 · Page 3, What changed since my last review

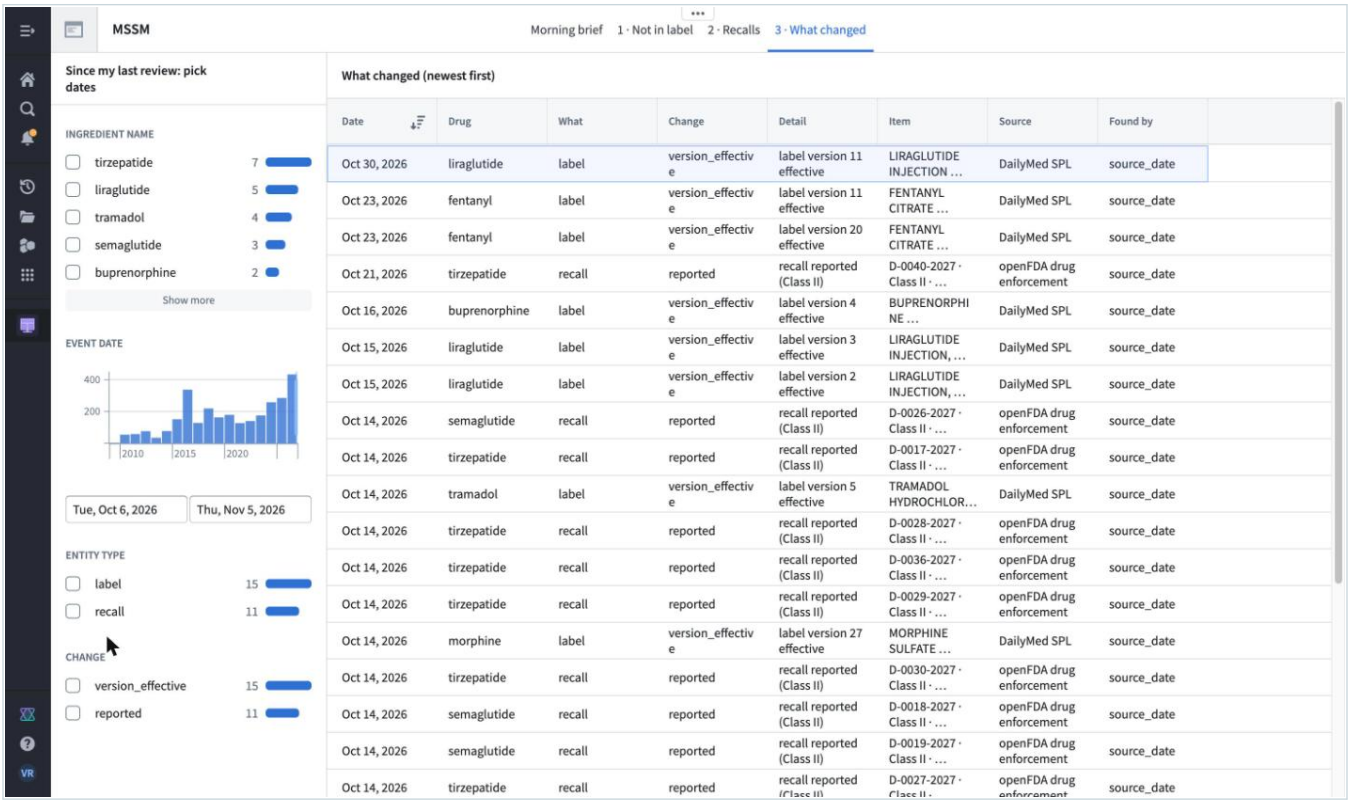


Figure 7. Page 3 with the start date set by the 'Past month' preset (6 Oct to 5 Nov 2026). Left: filters with counts. Right: the dated changes, newest first.
Source: docs/11-handin/screens/app-3b-what-changed-past-month.jpg (Workshop, 2026-11-05)

1. Click the tab *3 · What changed*.
2. Under *Event Date*, click the start-date box and pick the day of your last review (or a preset such as *Past month*). That is what 'since yesterday' means here: since *my* last look, because the sources do not all update daily (D045).
3. Check the count. With *Past month* the table holds 26 events, 15 label versions and 11 recalls (`screens/README.md`): the same number as card 3 on the Morning brief.
4. Narrow with the other filters: *Ingredient Name*, *Entity Type* (label, recall, signal) and *Change* (version effective, reported, terminated).
5. Read *Found by*: `source_date` means the source itself carries the date (a recall's report date, a label's effective date). `snapshot_diff` means we found it by comparing this build with the last one.

Gate: Morning routine TESTED

Required evidence: Each step works in View mode (as a user, not as the editor).

Result: Met: tabs switch pages, the drug filter narrows page 1, the labels panel lists the searched labels with working DailyMed links, recalls filter by linked drug, changes sort newest first, the cards read 1,073 · 126 · 26 (A016). Times (on v0.9.0): brief 11.5 s; switches 1.1, 0.8 and 2.4 s OBSERVED. In v0.10.0 the AI table lists the semaglutide quotes and the triage button keeps its height.

If it fails: If a card and its page disagree, the card's object set and the page's object set have drifted apart: they must start from the same set.

04

What each number means

Every number on the screen is a count of objects or a property of one object. Here is what each one means, and what it must not be read as.

One limit applies to all of them: FAERS counts **reports**, not patients or users, and reporting rises after news coverage, lawsuits or FDA warnings (stimulated reporting, DQ04, a known gap left to the reviewer). A high PRR says 'look here', not 'this is dangerous'.

Morning brief

Number	Definition	Do not read it as	Evidence
1,073 ranked signals	MSSM Signals with Is Ranked = true: clinical reactions that meet the Evans rule, are not found in any reference label, and come mostly from normal reports (not mass reports).	1,073 side effects caused by our drugs. It is a review queue.	DERIVED
126 active recalls	MSSM Recalls for our drugs with Is Active = true, which means openFDA status *Ongoing* (`clean_recalls.py` l.69), out of 900.	Recalls of products on our shelves: some match none of our products.	OBSERVED
26 changes in 30 days	MSSM Change Events whose event date is in the last 30 days, relative to today.	'Since yesterday', or new safety problems. Most are new label versions and recalls.	DERIVED

Page 1 signal record (selected rows)

Field on screen	Meaning	NAION value
Reports: this drug + reaction	a: current FAERS reports naming this drug as a suspect and this reaction	394
All reports for this drug	Every current report naming this drug as a suspect	20,892
PRR (x more often than other drugs)	Share of this reaction among this drug's reports, divided by its share among all other drugs' reports	112.5
PRR 95% CI low / high	The range the PRR is likely in, given the counts. A low bound well above 1 means the excess is not a small-numbers accident. It does not rule out reporting bias.	92.19 / 137.24
Chi² (Yates)	How surprising the count is. Evans rule (D006): 3+ reports, PRR 2+, chi² 4+	10653.8
Label status	not_in_label, possibly_in_label (found only through a medium synonym) or in_label	not_in_label
Reference labels searched	How many reference label versions of this drug were searched; the panel below lists them	3
Labels also read by AI	How many of those labels the AI also read (D041 v2)	3
Label check method	The rule versions in force (AI v1, text, synonyms v2); it does not say which check found what on this signal	text+ai_v1+syn_v2

Field on screen	Meaning	NAION value
Evaluation id	A fingerprint of the numbers above; a triage decision stores it. It changed for every semaglutide signal when the AI began reading their labels (no decision was on them)	c4b5da0b64d1e1e6
Bucket	Only clinical is ranked; efficacy, product use, abuse and others are tagged (D016 v3). Reports with 30+ drugs are set aside and counted separately (D023 v2)	clinical

Gate: Numbers explained TESTED

Required evidence: Each number on screen has a definition and a 'do not read it as'.

Result: Met in this chapter. On screen, the labels are plain but there is no help text yet.

If it fails: If the officer quotes '1,073 problems' to her committee, the screen needs a sub-title under the card that says 'to review'.

05

For editors: the widgets and how to change them

The whole module is built from six kinds of widget and a handful of shared variables. This chapter says what each one does for the officer and where an editor changes it.

Workshop term	Meaning
Module	One Workshop app. Ours: MSSM Safety Officer, in the project folder `Workshop/`.
Page	A screen inside the module. Ours are named 0 to 3.
Widget	A building block on a page: tabs, filter list, object table, property list, metric card, button.
Variable	A named value the widgets share: an object set, a selected object, a string, a count.
Object set	A list of Ontology objects, for example 'MSSM Signal where Is Ranked = true'.
Event	What a click does, for example 'switch to page'.
View mode / Edit mode	View mode is what users see. Edit mode is the builder: the same pages, plus a configuration panel for the selected widget and a list of variables.
Version	Editors save and publish numbered versions; users see the published one (v0.10.0 today).

The widgets on the screen

Widget	Where	What it does for her	How an editor changes it
Tabs	Header	Switches between the four pages	Select the tabs widget; each tab's on-click event is *Switch to page*. Page names carry a number because the editor sorts pages alphabetically (learning 2026-11-05)
Filter list	Left side of pages 1 to 3	Narrows the table by drug, active, class, date, type	Set its input to the **unfiltered** object set, pick the properties to filter on (or a linked object's property, as for Drug on page 2) and the display (pills, or a list with counts and a histogram)
Object table	Pages 0 to 3	Lists signals, recalls, changes, sources, decisions, labels searched, AI quotes	Set the input variable, the columns and their order, the default sort, and the row height (use *Absolute*: *Flex* collapses in a crowded section, learning 2026-11-04)
Property list	Right side of pages 1 and 2	Shows every field of the selected signal or recall	Set the input to the selected-object variable; choose fields, order and plain labels. Number format (decimals) is set on the Ontology property in Ontology Manager, so every app shares it
Metric card	Page 0	The three morning numbers	Each metric is a *Count* variable over an object set; change the set's filter to change what is counted
Button (Action)	Page 1	*Triage this signal* opens the decision form	Choose the Action, then set each field's default from a variable and lock it (Disabled). Who may submit is set on the Action type in the Ontology, not on the button

The variables behind them

Variable	Defined as	Read by
Ranked signals	MSSM Signal where Is Ranked = true	Drug filter, card 1
Ranked signals for chosen drugs	Ranked signals + the drug filter's output	Page 1 table
Selected signal key / evaluation id	Properties of the table's selected signal	Triage button defaults; decision list; labels panel
Decisions for selected signal	MSSM Triage Decision where Subject Key = Selected signal key	Decisions table
All recalls → Recalls for chosen drugs	MSSM Recall (900) → + the recall filters	Page 2 filter, then table
Active recalls	All recalls where Is Active = true	Card 2
All changes → Changes for chosen filters	MSSM Change Event (2,898) → + the change filters	Page 3 filter, then table
Changes last 30 days	All changes where Event Date is from 30 days ago, relative to today	Card 3
AI reactions quoted from labels searched	*Reference labels searched…* → MSSM Labeled Reaction (v0.10.0)	Page 1 AI table

Source for both tables: `foundry/ontology/README.md` (Workshop module) and A016.

How an editor makes a change, step by step

1. Open the module and switch to Edit mode (only editors of the project can).
2. Click the widget on the page. Its configuration opens on the side: input variable, columns or fields, sort, display.
3. To change *what* is shown everywhere at once, edit the **variable**, not the widget: every widget that reads it follows. Example: adding a filter to *Ranked signals* changes card 1, the drug filter and the page 1 table together.
4. For many columns at once, use the widget's JSON config (</> → edit → *Validate and save*), shown below.
5. Switch back to View mode and click through the page as the officer would.
6. Publish a new version. Users see it on their next load.

> **Where a change ripples**
> Two places, two scopes. A change on the **Ontology** (a property's display name or number format) shows in every app that uses that object type. A change on a **widget** affects that widget only.

Three patterns worth copying

Two object sets per page. An unfiltered set feeds the filter list; the filter's output makes a second, filtered set; that set feeds the table; the table's selected row feeds the detail panel. If the filter read the filtered set, it would feed back into itself, and every other drug would disappear from its own choices.

Cards count the same sets the pages show. Card 1 counts *Ranked signals*, the set behind page 1; card 2 counts *All recalls* where active; card 3 counts *All changes* in the last 30 days. So a card and its page cannot disagree.

The Action button with locked defaults. *Triage this signal* runs the Action *Create MSSM Triage Decision*. Subject Type is a static 'signal'; Subject Key and Evaluation Id come from the selected-signal variables. All three are locked, so a decision can only be saved against the exact signal and numbers on screen. The decision list refreshes after submit (D042).

The JSON config shortcut

Writing table columns and property-list fields straight into a widget's JSON replaced roughly ten clicks per column with one paste (learning 2026-11-05); reordering and relabelling the 22 signal properties took one paste instead of about 60 clicks (learning 2026-11-04). A column entry looks like this (the type id is in the object type's web address in Ontology Manager):

```
{"type": "property",
 "property": {"propertyType": {"objectTypeId": "<object type id>",
                               "propertyTypeId": "<column name>"}}}
```

Watch out

The JSON box is live: typing into it by accident edits the widget. Close it without *Validate and save* to discard (learning 2026-11-05).

Gate: Editor guide TESTED

Required evidence: Another editor can rebuild or change the module from this chapter and `foundry/ontology/README.md`.

Result: Met: every page's widgets, variables, filters, cards and Action defaults are written down. The module itself lives only in Foundry (no export in the repo).

If it fails: If a widget's configuration is lost, the README tables give its columns and order.

06

What we changed for usability, and why

Each change came from looking at the screen as the officer would. The trigger is written next to every change. 'Step' is a stage in our numbered build order: step 8 built the first screen, step 9 widened it to all 13 drugs, step 11 added the Morning brief, recalls and changes.

When	Before	Now	Why (her view)	Source
Step 8	PRR and chi² with 12 decimals	PRR 1 decimal, CI 2, chi² 1 (set on the Ontology property)	Precision she cannot use is noise	A012
Step 8	All signal properties in no useful order	22 rows in reading order with plain labels	Read top to bottom: why flagged, label check, trace	A012; README
Step 8	Past-decisions table collapsed (row height Flex)	Row height Absolute 160 px; columns time · decision · reviewer · rationale · evaluation id	She must see her history	Learning 2026-11-04; README
Step 9	One ranked list mixing 12 drugs (exenatide has no reference label, so no signals; R07)	Drug filter pill with counts per drug	She reviews one drug at a time	A012 update
Step 11	One page: question 1 only	Morning brief plus one page per question	All three questions, from one place	A016
Step 11	No freshness anywhere	'Data as of' table per source	'Nothing changed' needs a date	A016, D045
Step 11	Pull time blank for FAERS and RxNorm	FAERS: 'quarterly file (no pull time)'; RxNorm: the pull time	A blank cell looks like a fault	`ontology_changes.py` l.152, l.158
Step 11	Recall class as a code	"Class (I = most serious)"	No lookup needed	A016
Step 11	'Since yesterday' undefined	Start date on page 3 = since my last review	Sources are not daily	D045
Step 11	Record showed only how many labels were searched	Panel *Reference labels searched* with version, date, sections and DailyMed link	"Show me where it says that"	A016
6 Nov	AI quotes only in Object Explorer; AI read 2 oxycodone labels	Page 1 table *What the AI read in these labels* for the selected drug; AI also reads the 3 semaglutide labels (v0.10.0)	"What does the label say about the eyes?"	A019, D041 v2
6 Nov	A fixed-height table squeezed the column; *Triage this signal* was cut off	*Enable scrolling* on the Evidence and triage section	The button must always be there	Learning 2026-11-06

The ranking itself also changed for her sake. Non-harm terms (efficacy complaints, product handling, procedures) moved out of the ranked queue into tagged buckets, and a reviewed synonym table removed label wording misses. The queue went from 1,355 at step 9 (first full build) to 1,210 (synonym table v1), 1,115 (wider buckets, D016 v3) and 1,073 (synonym table v2). R05 explains each step **OBSERVED**.

Gate: Usability changes TESTED

Required evidence: Each change has a before, an after and a reason tied to the officer.

Result: Met for the twelve changes above. None was tested with a real officer.

If it fails: If a change cannot be tied to one of her questions or the trust rule, revert it.

07

Known gaps and the next UI improvements

What a careful reviewer will notice on the screen today, and what we would build next.

Known gaps

Gap	Where	Effect	Fix
Human 'morning in two minutes' walk-through not run (machine timings done)	all	Her reading time is unmeasured	Run it with the safety-officer persona, then a real officer
Only the owner can submit a decision	page 1	The officer cannot use the write-back yet	Grant the officer role on the Action
AI quotes for two drugs only	page 1	The AI reads 5 labels (oxycodone and semaglutide); for the other drugs the AI table is empty	Widen the AI after a pharmacist checks the key
Reviewer shows a user ID, not a name	page 1 decisions	Harder to read the history	Show the user's display name
No help text ('Matched to our products: 0', what the cards count)	pages 0, 2	A compounded-product recall looks like a bug; a card can be read as findings	One line of help text under each card and field
No dedicated Signal → Label link type	page 1	None for her: the panel walks existing links. A builder cannot ask 'which signals did this label clear?' directly	A join dataset and a link type
Signal changes only appear after the next FAERS quarter	page 3	Today's feed is labels and recalls; the build-to-build diff found 0 changes on unchanged data	Expected; say so on the page

Next UI improvements

None of these is built. They are listed in the order we would build them.

Improvement	Why (her view)	What it needs	Status
Help text under each card and tricky field	'1,073 to review', not '1,073 problems'; 'Matched 0' explained	Text only, in the widgets	PLANNED
Officer permission on the triage Action	She can record her own decisions	A role for the officer on the Action type	PLANNED
Reviewer display name	She can read the history at a glance	Show the user's name, not the ID	PLANNED
'Evidence changed since your decision' flag	A decision made on old numbers should be re-looked at	Compare the decision's evaluation id with the signal's current one (both are on screen today)	PLANNED
AI quotes for the other drugs	'Show me where it says that' for every drug, not only oxycodone and semaglutide	The AI reads the other drugs' reference labels (after the pharmacist check)	PLANNED

Improvement	Why (her view)	What it needs	Status
Health page	She (and IT) see a stale source or a failed check before trusting the numbers	A key column and an object type for `monitoring_checks` (21 checks per build)	PLANNED
Per-user 'last reviewed'	Page 3 opens at her last review; the brief counts 'new since you last looked'	Per-user state, or the time of her latest triage decision	PLANNED
Triage on recalls	Recalls need decisions too	A Triage Decision → Recall link	PLANNED

Gate: Ready for a pilot user PLANNED

Required evidence: A real officer completes the morning routine unaided in two minutes and can trace a flag to its source.

Result: Not yet: the module answers all three questions and the machine timings are fast (v0.9.0: brief 11.5 s, switches 0.8 to 2.4 s), but the human timed test and the real-user test are still to do, and she cannot submit a decision yet.

If it fails: Fix the step where she stops, before adding any new feature.

A

Reference

Figures quoted from project records (not in numbers.json)

These come from project records, not from the baseline-gate capture. Each is also cited where it appears.

Figure	Source
About fifteen minutes each morning; persona quotes	docs/02-design-thinking/design-thinking-canvas.md (fictional persona)
Ozempic v20 (2026-06-01), Wegovy v19 (2026-06-18), oral Ozempic v14 (2026-01-30)	A016; screenshot app-1c-labels-searched-naion.jpg
26 events = 15 label versions + 11 recalls (Past month)	screens/README.md; app-3b screenshot
22 signal properties; 15 recall fields	foundry/ontology/README.md (Workshop module)
PRR 1 decimal, CI 2, chi² 1 (was 12 decimals)	A012; foundry/ontology/README.md (Polish, v0.3)
Decisions table row height Absolute 160 px; Flex collapses	foundry/ontology/README.md; learning 2026-11-04
About 10 clicks per column; about 60 clicks for 22 properties	Learnings 2026-11-05 and 2026-11-04
Evans rule: 3+ reports, PRR 2+, chi² 4+	D006
Mass reports: 30+ drugs per report	D023 v2
AI found diabetic-retinopathy terms, no optic-nerve term (NAION)	A019; screenshot app-1e-ai-quotes-panel-naion.jpg
Triage button cut off by a fixed-height table; fixed with Enable scrolling	DECISIONS learning 2026-11-06; A019

Numbers used in this report

Every number in the table below is looked up from `docs/11-handin/numbers.json` at build time. Source: Foundry SQL (repo SQL panel) after the baseline-gate rebuild of the whole downstream chain: raw_dailymed_spl → clean_labels → clean_label_product_map → ontology_objects → ai_label_extraction → ontology_signals → ontology_changes → monitoring. Build durations from the Foundry Builds page. Query: docs/11-handin/baseline-gate.md. Captured: 2026-11-05T07:50Z. Figures quoted from other project records (decision log, data-issues register, optimization log) name that record where they appear.

Key	Value
`ai.grounded`	745
`ai.pilot_labels`	5
`ai.reactions`	745
`app.ai_quotes_panel_rows_naion`	297
`app.labels_read_by_ai_naion`	3
`app.labels_searched_naion`	3
`app.pages`	4

Key	Value
app.timing_s.morning_brief_load	11.5
app.timing_s.switch_recalls	1.1
app.timing_s.switch_signals_with_labels	2.4
app.timing_s.switch_what_changed	0.8
app.workshop_version	v0.10.0
changes.last_30_days	26
changes.snapshot_diff_after_rebuild	0
changes.total	2,898
monitoring.checks_per_build	21
monitoring.faers_age_days	130
recalls.active	126
recalls.newest	D-0040-2027
recalls.newest_reported	2026-10-21
recalls.total	900
scope.drugs_formulary	13
scope.drugs_with_signals	12
scope.glp1	5
scope.opioids	8
signals.naion.a	394
signals.naion.all_reports_drug	20,892
signals.naion.chi2_yates	10653.8
signals.naion.drug	semaglutide
signals.naion.evaluation_id_v2	c4b5da0b64d1e1e6
signals.naion.prr	112.5
signals.naion.prr_ci_high	137.24
signals.naion.prr_ci_low	92.19
signals.naion.reaction	Optic ischaemic neuropathy
signals.possibly_in_label	108
signals.ranked	1,073
signals.ranked_history.d016_v3	1,115
signals.ranked_history.step9	1,355

Key	Value
signals.ranked_history.syn_v1	1,210
signals.ranked_history.syn_v2	1,073